Te Lai

Email: tl243@duke.edu
Shenzhen, Guangdong, China

Research Interests

Environmental & Energy Economics, Climate Policy, Clean Energy & Electric Vehicle (EV) Supply Chains, Carbon Pricing & Markets, Big Data Applications in Applied Microeconomics

Education

University of Michigan, Ann Arbor — **Ann Arbor, MI, USA**

Master of Science in Applied Economics — 08/2022 – 12/2023

- GPA: 4.0 / 4.0
- Selected Coursework: Python for Data Science, Business Data Analytics, Advanced Econometrics (focused on causal inference and panel data methods), Financial Modeling, Valuation Analysis

Duke Kunshan University (Partner of Duke University) — **Kunshan, Jiangsu, China**

Bachelor of Arts in Political Economy — 08/2018 – 05/2022

- GPA: 3.8 / 4.0 (Honors: Fall 2020 Dean's List)
- Selected Coursework: Probability and Statistics, Calculus, Multivariable Calculus, Intermediate Microeconomics, Intermediate Macroeconomics, Econometrics

Research Experience

Duke Kunshan University — **Kunshan, China**

Research Assistant in Environmental Economics (Advisor: Prof. Jingbo Cui) — 02/2024 – Present

- **Co-authored Paper:** Co-authored the working paper "Global Carbon Pricing and Clean Energy Innovation", employing a staggered difference-in-differences (DID) and Poisson pseudo-maximum likelihood (PPML) framework to estimate the causal effects of carbon taxes and Emissions Trading Schemes (ETS) on global clean energy innovation.
- **Green Tech Classification via LLMs:** Led the integration of LLM APIs (DeepSeek, etc.) into academic research pipelines. Designed prompting and verification frameworks to classify 5M+ international patents into 500+ green technology categories, including renewable energy development, carbon accounting, and industrial decarbonization pathways, achieving an 80% classification accuracy (significantly outperforming the International Energy Agency's baseline of 25%).
- **Clean Energy & EV Supply Chain Tracking:** Curated, cleaned, and merged multi-million-scale global patent records (Lens, USPTO, EPO) to track supply chains of clean energy and transportation electrification (EV/battery technologies), providing a robust empirical foundation for research on green innovation.
- **Econometric Analysis & Empirical Support:** Formulated empirical strategies, compiled data indicators, and conducted econometric analysis to support team papers on green innovation, carbon markets, and supply chain resilience.

Duke Kunshan University **Kunshan, China**

Research Assistant (Advisor: Prof. Jingbo Cui) 05/2021 – 03/2022

- **Data Processing & Panel Econometrics:** Cleaned and organized over 500,000 patent records and financial data of 5,000+ Chinese listed companies using Python and Stata, executing econometric models to analyze network spillovers.
- **Social Network Analysis:** Quantified the centrality of collaborative patent networks using social network analysis in Python. Conducted econometric analysis in Stata to evaluate the impact of network position on firm performance and green technology adoption, identifying a 0.3% excess return associated with network centrality.
- **Co-authored Paper:** Co-authored the manuscript titled "The Evolution of The Network: The Evidence from China", representing the first comprehensive visualization study of national patent networks in China.
- **Conference Presentation:** Presented findings at the 6th East Asian Association of Environmental and Resource Economics (AAERE) Conference hosted by Seoul National University.

Duke Kunshan University **Kunshan, China**

Research Assistant (Advisor: Prof. Jingbo Cui) 09/2020 – 12/2020

- **Data Engineering:** Processed and analyzed a database of over 10 million overseas (European and American) patent records using Python, with a specific focus on green technologies.
- **Quality Comparison:** Conducted comparative analysis on patent quality differences between Sino-foreign collaborative patents and single-country application patents, providing analytical support for international collaborative R&D policy evaluations.

Duke Kunshan University **Kunshan, China**

Research Assistant (Advisor: Prof. Jingbo Cui) 01/2020 – 04/2020

- **Patent Classification:** Collected and classified environmental protection patents filed in China, building database metrics including patent status, applicant type, application date, and citation info.
- **Industry Analysis:** Compared the quantity and quality of environmental protection patents across different industries to assess sectoral responses to green incentives.
- **Policy Review & Literature Synthesis:** Conducted literature reviews on the impact of environmental policies (e.g., carbon reduction targets, renewable portfolio standards) on the development and patenting of the environmental protection industry.

Working Papers

Cui, Jingbo, Te Lai, and Yang Zheng. "Global Carbon Pricing and Clean Energy Innovation." (Working Paper, 2026).

Cui, Jingbo, and Te Lai. "The Evolution of The Network: The Evidence from China." (Under review / Working Paper).

Conference Presentations

"The Evolution of The Network: The Evidence from China", The 6th East Asian Association of Environmental and Resource Economics (AAERE) Conference, Seoul National University, South Korea, August 2021.

Skills & Credentials

- **Programming:** Python (Pandas, NumPy, Scikit-learn, PyTorch, Selenium, ccxt), R, Stata, SQL, LaTeX, Git
- **Data & Tools:** Large-Scale Databases (Lens, USPTO, EPO, Orbis), Bloomberg, Wind
- **Languages:** Mandarin Chinese (Native), English (Fluent)
- **Credentials:** CFA Level 3 Candidate, Financial Risk Manager (FRM) Level 1